

PicoCTF 2026 – Forensics Git 2

Forensics Git 2

Forensics Medium 400 pts

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The agents interrupted the perpetrator's disk deletion routine. Can you recover this git repo?

Download the disk image [here](#) ↗.

In this challenge, they want us to “recover a Git repo”, but what does this mean?

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Extracting the Filesystem

```
└─$ tsk_recover -f ext4 -e -o 1140736 -i raw -v disk.img .  
tsk_img_open: Type: 1   NumImg: 1   Img1: disk.img  
tsk_img_findFiles: disk.img found  
tsk_img_findFiles: 1 total segments found
```

Like the previous challenges in this series, our first task is to extract the filesystem from the disk image so we can interact with the Git repo

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Extracting the Filesystem

```
└─$ tsk_recover -f ext4 -e -o 1140736 -i raw -v disk.img .  
tsk_img_open: Type: 1   NumImg: 1   Img1: disk.img  
tsk_img_findFiles: disk.img found  
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```

We do this using The Sleuth Kit's `tsk_recover` tool

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Extracting the Filesystem

```
└─$ find . -name ".git"
./home/ctf-player/Code/killer-chat-app/.git

└─(theabyss@hackerforge)-[/tmp/hf/f_git2]
└─$ cd ./home/ctf-player/Code/killer-chat-app/
```

Afterwards, we locate the Git repo using the `find` command, then enter that directory

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Recovering the Git Repo

```
└─$ git log -p  
fatal: not a git repository (or any parent up to mount point /)  
Stopping at filesystem boundary (GIT_DISCOVERY_ACROSS_FILESYSTEM not set).
```

When we attempt to get the commit history using the Git `log` command, we get a specific error

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Recovering the Git Repo

```
└─$ git log -p  
fatal: not a git repository (or any parent up to mount point /)  
Stopping at filesystem boundary (GIT_DISCOVERY_ACROSS_FILESYSTEM not set).
```

This error occurs because the Git repo has been partially deleted

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Recovering the Git Repo

```
└─$ git log -p  
fatal: not a git repository (or any parent up to mount point /)  
Stopping at filesystem boundary (GIT_DISCOVERY_ACROSS_FILESYSTEM not set).
```

With a bit of research, we discover that the `.git/refs/heads` and `.git/refs/tags` directories are missing

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Recovering the Git Repo

```
mkdir -p .git/refs/heads && mkdir -p .git/refs/tags
```

So we create those directories and try again

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Recovering the Git Repo

```
└─$ git log -p  
fatal: your current branch 'master' does not have any commits yet
```

But now we get a different error: we don't have any commits!

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Recovering the Git Repo

```
└─$ tail -1 .git/logs/refs/heads/master | cut -d' ' -f2  
01533f718556a0e59f1467dae4fa462eed82c2a1
```

We can restore the commit history in two steps.
The first step is to retrieve the last commit hash
from the logs directory

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Recovering the Git Repo

```
git update-ref refs/heads/master  
01533f718556a0e59f1467dae4fa462eed82c  
2a1
```

The second step is to use this commit hash with the Git **update-ref** command

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Recovering the Git Repo

```
diff --git a/logs/3.txt b/logs/3.txt
deleted file mode 100644
index 7178644..0000000
— a/logs/3.txt
+++ /dev/null
@@ -1,3 +0,0 @@
-Rex: Meet at the old arcade basement for the secret hideout.
-Jay: Ask Rusty at the door and use password picoCTF{p17_r35t03_10000bf3}.
-Rex: Bring the decoder map so we can plan the route.
```

Finally, we can get the commit history from the Git repo, which contains the flag